

The screenshot shows a web browser window displaying the HashiCorp Terraform documentation page for Google Cloud. The browser's address bar shows the URL `developer.hashicorp.com/terraform/tutorials/gcp-get-started`. The page has a navigation bar with links for Terraform, Install, Tutorials, Documentation, Sandbox, Registry, and Try Cloud. A search bar is also present. On the left, a sidebar lists various topics, with 'GCP' highlighted under the 'Tutorials' section. The main content area is titled 'Get Started - Google Cloud' and includes a sub-header 'Build, change, and destroy Google Cloud Platform (GCP) infrastructure using Terraform. Step-by-step, command-line tutorials will walk you through the Terraform basics for the first time.' Below this, there are three tutorial cards, each marked as 'Complete'. The first card is titled 'What is Infrastructure as Code with Terraform?' and describes learning how to safely build, change, and manage infrastructure. The second card is titled 'Install Terraform' and describes installing Terraform on Mac, Linux, or Windows, creating a Docker container, and following a quick-start tutorial. The third card is partially visible and also marked as 'Complete'. The Windows taskbar at the bottom shows the time as 17:52 on 10-01-2026.

developer.hashicorp.com/terraform/tutorials/gcp-get-started

Terraform Install Tutorials Documentation Sandbox Registry Try Cloud

Search

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Get Started - Google Cloud

Build, change, and destroy Google Cloud Platform (GCP) infrastructure using Terraform. Step-by-step, command-line tutorials will walk you through the Terraform basics for the first time.

Review Complete

Complete

What is Infrastructure as Code with Terraform?

Learn how infrastructure as code lets you safely build, change, and manage infrastructure. Try Terraform.

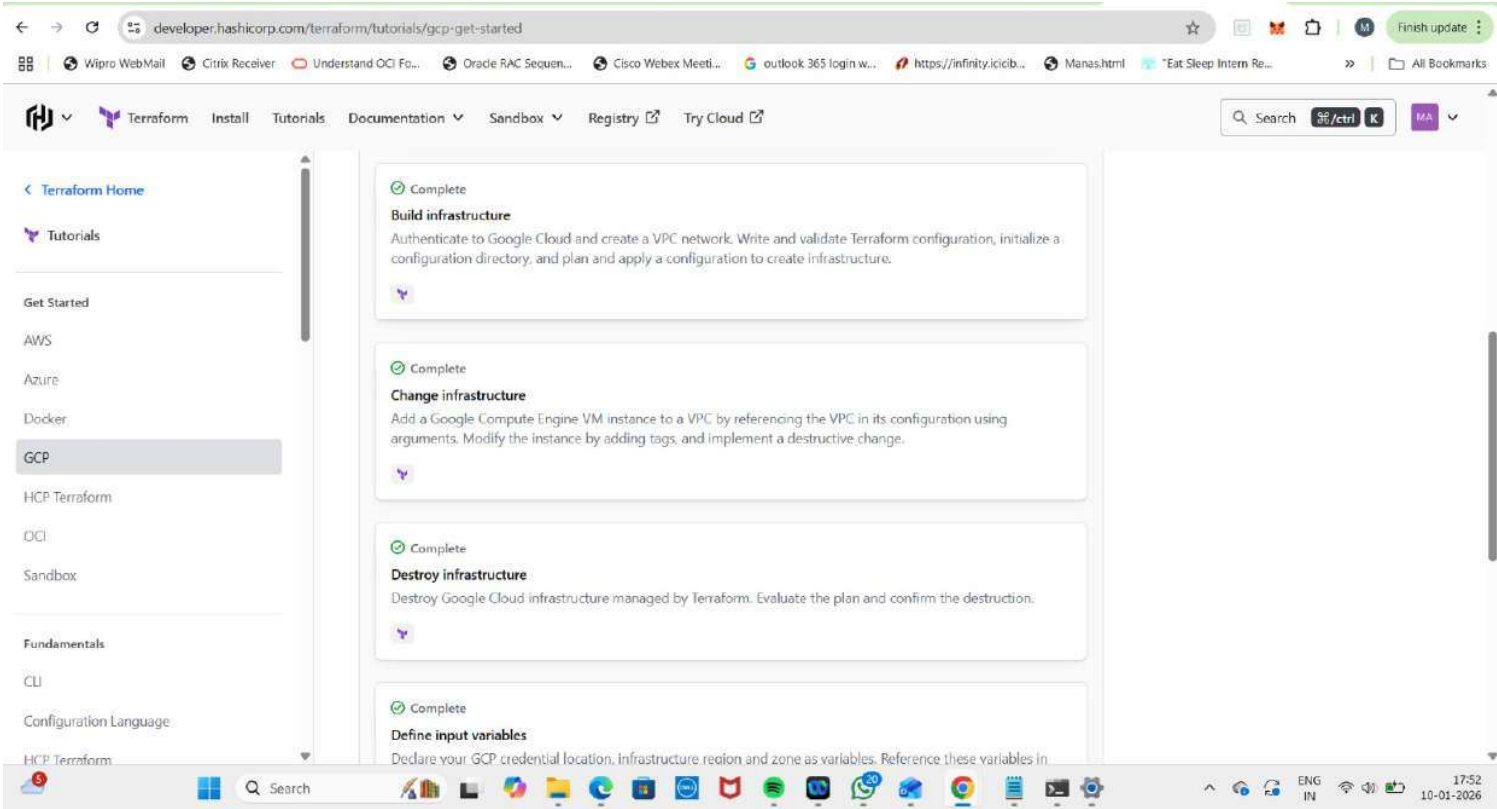
Complete

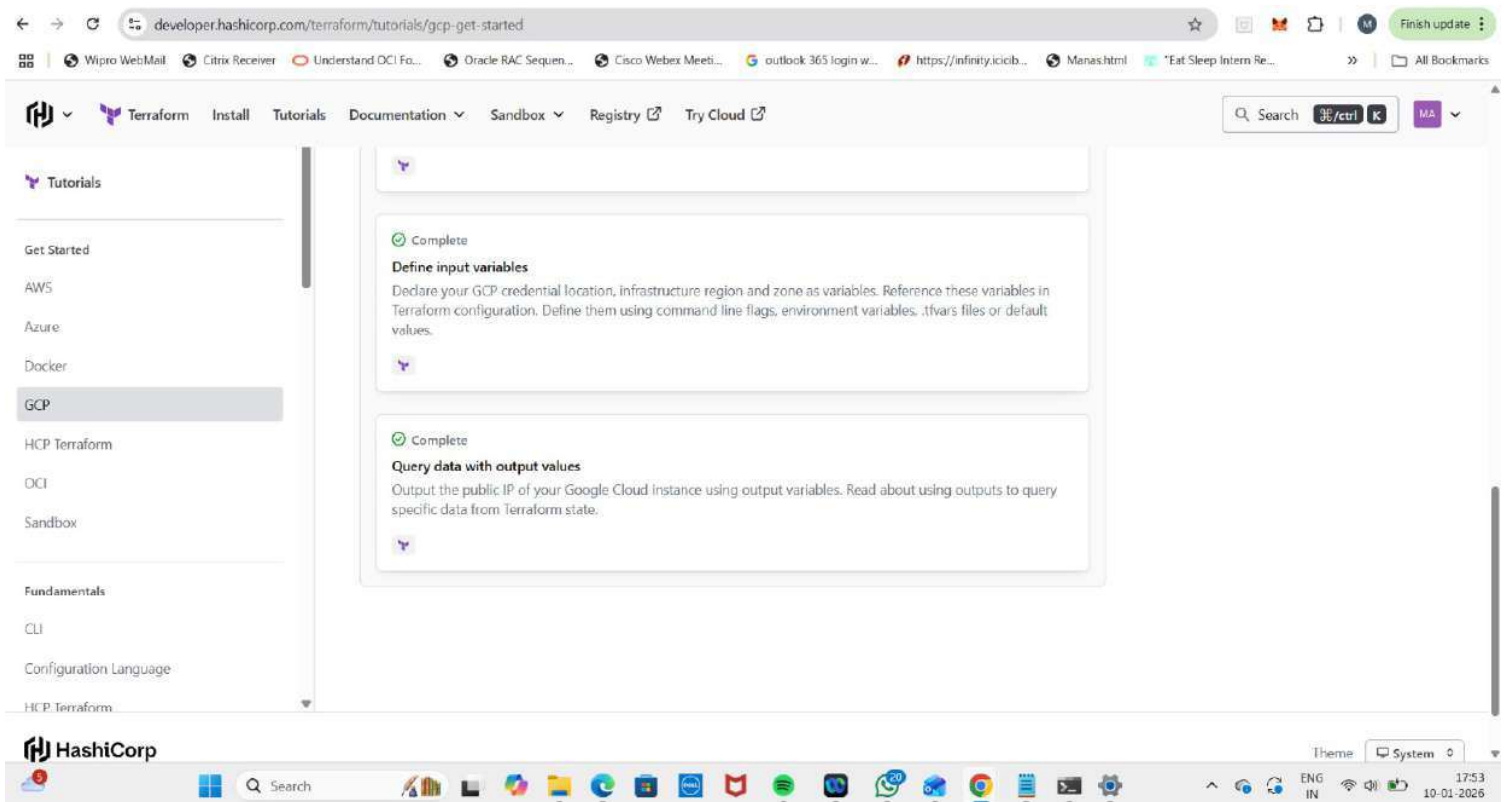
Install Terraform

Install Terraform on Mac, Linux, or Windows by downloading the binary or using a package manager (Homebrew or Chocolatey). Then create a Docker container locally by following a quick-start tutorial to check that Terraform installed correctly.

Complete

17:52 10-01-2026





Terraform Fundamentals

skills.google/course_templates/636/labs/592696

Wipro WebMail | Citrix Receiver | Understand OCI Fo... | Oracle RAC Sequen... | Cisco Webex Meeti... | outlook 365 login w... | https://infinity.cicib... | Menas.html | *Eat Sleep Intern Re... | All Bookmarks

Google Skills

What do you want to learn today?

140 0 ? M

Build Infrastructure with Terraform on Google Cloud > Terraform Fundamentals

Apply your skills in Google Cloud console [Get started](#)

Contents

Terraform Fundamentals

[Send feedback](#)

[End Lab](#) 00:13:43 Time limit

Caution: When you are in the console, do not deviate from the lab instructions. Doing so may cause your account to be blocked. [Learn more.](#)

[Open Google Cloud console](#)

Username
student-b2-1d688e5e7d57f

Password
j2NRNcAJR1XV

Project ID

[Lab](#) 10 minutes 1 Credit Introductory

★★★★★

This lab may incorporate AI tools to support your learning.

Terraform Fundamentals

GSP156

Google Cloud Self-Paced Labs

Objectives 100/100

- Setup and requirements
- What is Terraform?
- Task 1. Verify a Terraform installation
- Task 2. Build the infrastructure
- Task 3. Test your understanding
- Congratulations!

[Previous](#) [Next](#)

Temps to rise Next Monday

Search

ENG IN 15:41 09-01-2026

Infrastructure as Code with Terraform

The screenshot shows the Google Skills interface for a lab titled "Infrastructure as Code with Terraform". The page is part of a course "Build Infrastructure with Terraform on Google Cloud". The lab title is "Infrastructure as Code with Terraform". The lab has a time limit of 00:29:54 and is rated 5 stars. It is labeled as a "Lab" and takes 25 minutes, worth 5 credits, at an intermediate level. The lab description states: "This lab was developed with our partner, Hashicorp. Your personal information may be shared with Hashicorp, the lab sponsor, if you have opted-in to receive product updates, announcements, and offers in your Account Profile." The lab instructions and tasks are listed on the right: "Lab instructions and tasks", "GSP750", "Overview", "Setup and requirements", "Task 1. Build infrastructure", "Task 2. Change the infrastructure", "Task 3. Create resource dependencies", "Task 4. Provision infrastructure". The left sidebar shows navigation options: "Dashboard", "Catalog", "Paths", "Collections", and "Subscriptions". The bottom of the page shows a Windows taskbar with various application icons and the system clock indicating 19:52 on 09-01-2025.

skills.google/course_templates/636/labs/592697

Google Skills

What do you want to learn today?

Build Infrastructure with Terraform on Google Cloud > Infrastructure as Code with Terraform

Apply your skills in Google Cloud console [Get started](#)

Contents

Infrastructure as Code with Terraform

End Lab 00:29:54 Time limit

Caution: When you are in the console, do not deviate from the lab instructions. Doing so may cause your account to be blocked. [Learn more](#)

[Open Google Cloud console](#)

Username

student-02-a85298d4db9af

Password

81DmeXPTxjWG

Project ID

Infrastructure as Code with Terraform

Lab 25 minutes 5 Credits Intermediate

★★★★★

This lab may incorporate AI tools to support your learning.

This lab was developed with our partner, Hashicorp. Your personal information may be shared with Hashicorp, the lab sponsor, if you have opted-in to receive product updates, announcements, and offers in your Account Profile.

[Send feedback](#)

100/100

Lab instructions and tasks

GSP750

Overview

Setup and requirements

Task 1. Build infrastructure

Task 2. Change the infrastructure

Task 3. Create resource dependencies

Task 4. Provision infrastructure

[Previous](#) [Next](#)

Search

ENG IN 19:52 09-01-2025

Interact with Terraform Modules

The screenshot shows the Google Skills interface for a lab titled "Interact with Terraform Modules". The browser address bar shows the URL: `skills.google/course_templates/536/labs/592698`. The Google Skills header includes a search bar with the text "What do you want to learn today?", a share icon, a star rating of 140, a flame icon with 0, a question mark icon, a globe icon, and a user profile icon labeled "M".

The lab page has a breadcrumb trail: `Build Infrastructure with Terraform on Google Cloud > Interact with Terraform Modules`. On the right, there is a button to "Apply your skills in Google Cloud console" and a "Get started" button. A "Send feedback" button is also present.

The lab details section includes a timer showing "00:08:25" with a "Time limit" icon, a red "End Lab" button, and a caution note: "Caution: When you are in the console, do not deviate from the lab instructions. Doing so may cause your account to be blocked. Learn more." Below this is a button to "Open Google Cloud console".

The lab title "Interact with Terraform Modules" is prominently displayed. Below it, the lab is identified as a "Lab" taking "15 minutes" and worth "5 Credits", with an "Intermediate" difficulty level. It has a 5-star rating. A blue box contains the text: "This lab may incorporate AI tools to support your learning." Below this, a disclaimer states: "This lab was developed with our partner, Hashicorp. Your personal information may be shared with Hashicorp, the lab sponsor, if you have opted in to receive product updates, announcements, and offers in your Account Profile."

On the right side, a sidebar shows the lab's progress for user "GSP761", who has completed 100/100. The sidebar lists the following sections: Overview, Setup and requirements, What is a Terraform module?, Task 1. Use modules from the Registry, Task 2. Build a module, and Congratulations!

The bottom of the page features a navigation bar with "Previous" and "Next" buttons. The system tray at the very bottom shows the date and time as "09-01-2026 17:48" and the language as "ENG IN".

Tree SS

```
C:\Users\Manas\Documents\exercise5>tree /F
Folder PATH listing for volume OS
Volume serial number is 1A53-0AEC
C:..
.
├── .gitignore
├── .terraform.lock.hcl
├── curl
├── main.tf
├── outputs.tf
├── providers.tf
├── README.md
├── terraform.tfstate
├── terraform.tfstate.backup
├── terraform.tfvars
├── variables.tf
├── .terraform
│   ├── terraform.tfstate
│   ├── modules
│   │   └── modules.json
│   └── providers
│       ├── registry.terraform.io
│       │   ├── hashicorp
│       │   │   ├── archive
│       │   │   │   ├── 2.7.1
│       │   │   │   │   ├── windows_amd64
│       │   │   │   │   │   ├── LICENSE.txt
│       │   │   │   │   │   └── terraform-provider-archive_v2.7.1_x5.exe
│       │   │   │   └── google
│       │   │   │       ├── 5.45.2
│       │   │   │       │   ├── windows_amd64
│       │   │   │       │   │   ├── LICENSE.txt
│       │   │   │       │   │   └── terraform-provider-google_v5.45.2_x5.exe
│       │   └── google
│       └── google
│           ├── 5.45.2
│           │   ├── windows_amd64
│           │   │   ├── LICENSE.txt
│           │   │   └── terraform-provider-google_v5.45.2_x5.exe
│           └── google
└── modules
    ├── compute
    │   ├── function.zip
    │   └── main.tf
```

```
├── google
│   ├── 5.45.2
│   │   └── windows_amd64
│   │       ├── LICENSE.txt
│   │       └── terraform-provider-google_v5.45.2_x5.exe
└── modules
    ├── compute
    │   ├── function.zip
    │   ├── main.tf
    │   ├── outputs.tf
    │   └── variables.tf
    ├── frontend
    │   └── index.html
    ├── function
    │   ├── main.py
    │   └── requirements.txt
    ├── iam
    │   ├── main.tf
    │   ├── outputs.tf
    │   └── variables.tf
    ├── networking
    │   ├── main.tf
    │   ├── outputs.tf
    │   └── variables.tf
    ├── storage
    │   ├── main.tf
    │   ├── outputs.tf
    │   └── variables.tf
    └── frontend
        └── index.html
```

C:\Users\Manas\Documents\exercise5>gcloud config list project|

Terraform init and validate

```
C:\Users\Manas\Documents\exercise5>terraform init
Initializing the backend...
Initializing modules...
Initializing provider plugins...
- Reusing previous version of hashicorp/google from the dependency lock file
- Reusing previous version of hashicorp/archive from the dependency lock file
- Using previously-installed hashicorp/google v5.45.2
- Using previously-installed hashicorp/archive v2.7.1

Terraform has been successfully initialized!

You may now begin working with Terraform. Try running "terraform plan" to see
any changes that are required for your infrastructure. All Terraform commands
should now work.

If you ever set or change modules or backend configuration for Terraform,
rerun this command to reinitialize your working directory. If you forget, other
commands will detect it and remind you to do so if necessary.

C:\Users\Manas\Documents\exercise5>terraform validate
Success! The configuration is valid.

C:\Users\Manas\Documents\exercise5>
```



Terraform plan

should now work.

If you ever set or change modules or backend configuration for Terraform, rerun this command to reinitialize your working directory. If you forget, other commands will detect it and remind you to do so if necessary.

```
C:\Users\Manas\Documents\exercise5>terraform validate
Success! The configuration is valid.
```

```
C:\Users\Manas\Documents\exercise5>terraform plan
module.compute.data.archive_file.function_zip: Reading...
module.compute.data.archive_file.function_zip: Read complete after 0s [id=7c630c88d82de1702e1bd6ef7b41b19a2dcd18e1]
google_project_service.required_apis["run.googleapis.com"]: Refreshing state... [id=cloud-exercise-03-manassamant/run.googleapis.com]
google_project_service.required_apis["firestore.googleapis.com"]: Refreshing state... [id=cloud-exercise-03-manassamant/firestore.googleapis.com]
module.iam.google_service_account.cloud_function_sa: Refreshing state... [id=projects/cloud-exercise-03-manassamant/serviceAccounts/cloud-function-sa@cloud-exercise-03-manassamant.iam.gserviceaccount.com]
module.iam.google_service_account.cloud_run_sa: Refreshing state... [id=projects/cloud-exercise-03-manassamant/serviceAccounts/cloud-run-sa@cloud-exercise-03-manassamant.iam.gserviceaccount.com]
google_project_service.required_apis["compute.googleapis.com"]: Refreshing state... [id=cloud-exercise-03-manassamant/compute.googleapis.com]
google_project_service.required_apis["artifactregistry.googleapis.com"]: Refreshing state... [id=cloud-exercise-03-manassamant/artifactregistry.googleapis.com]
google_project_service.required_apis["cloudfunctions.googleapis.com"]: Refreshing state... [id=cloud-exercise-03-manassamant/cloudfunctions.googleapis.com]
google_project_service.required_apis["cloudbuild.googleapis.com"]: Refreshing state... [id=cloud-exercise-03-manassamant/cloudbuild.googleapis.com]
google_project_service.required_apis["vpcaccess.googleapis.com"]: Refreshing state... [id=cloud-exercise-03-manassamant/vpcaccess.googleapis.com]
module.storage.google_storage_bucket.frontend: Refreshing state... [id=cloud-exercise-03-manassamant-frontend]
module.networking.google_compute_network.vpc: Refreshing state... [id=projects/cloud-exercise-03-manassamant/global/networks/exercise5-vpc]
module.storage.google_storage_bucket.terraform_state: Refreshing state... [id=cloud-exercise-03-manassamant-tf-state]
module.compute.google_storage_bucket.function_source: Refreshing state... [id=cloud-exercise-03-manassamant-function-source]
module.iam.google_project_iam_member.cf_build_editor: Refreshing state... [id=cloud-exercise-03-manassamant/roles/cloudbuild.builds.builder/serviceAccount:cloud-function-sa@cloud-exercise-03-manassamant.iam.gserviceaccount.com]
module.iam.google_project_iam_member.cf_service_agent: Refreshing state... [id=cloud-exercise-03-manassamant/roles/cloudfunctions.serviceAgent/serviceAccount:cloud-function-sa@cloud-exercise-03-manassamant.iam.gserviceaccount.com]
module.iam.google_project_iam_member.cf_storage_access: Refreshing state... [id=cloud-exercise-03-manassamant/roles/storage.objectViewer/serviceAccount:cloud-function-sa@cloud-exercise-03-manassamant.iam.gserviceaccount.com]
module.iam.google_project_iam_member.cf_firestore_access: Refreshing state... [id=cloud-exercise-03-manassamant/roles/datastore.user/serviceAccount:cloud-function-sa@cloud-exercise-03-manassamant.iam.gserviceaccount.com]
module.storage.google_storage_bucket.iam_member.frontend_public_read: Refreshing state... [id=b/cloud-exercise-03-manassamant-frontend/roles/storage.objectViewer/allUsers]
module.storage.google_storage_bucket_object.frontend_files["index.html"]: Refreshing state... [id=cloud-exercise-03-manassamant-frontend-index.html]
module.iam.google_project_iam_member.cr_invoker: Refreshing state... [id=cloud-exercise-03-manassamant/roles/run.invoker/serviceAccount:cloud-run-sa@cloud-
```

Precip coming
In about 1.5 hours



Search



13:43
12-01-2026

```
module.iam.google_project_iam_member.cf_storage_access: Refreshing state... [id=cloud-exercise-03-manassamant/roles/storage.objectViewer/serviceAccount:cloud-function-sa@cloud-exercise-03-manassamant.iam.gserviceaccount.com]
module.iam.google_project_iam_member.cf_firestore_access: Refreshing state... [id=cloud-exercise-03-manassamant/roles/datastore.user/serviceAccount:cloud-function-sa@cloud-exercise-03-manassamant.iam.gserviceaccount.com]
```

Terraform used the selected providers to generate the following execution plan. Resource actions are indicated with the following symbols:
+ create

Terraform will perform the following actions:

```
# module.iam.google_project_iam_member.cr_invoker will be created
+ resource "google_project_iam_member" "cr_invoker" {
  + etag      = (known after apply)
  + id        = (known after apply)
  + member    = "serviceAccount:cloud-run-sa@cloud-exercise-03-manassamant.iam.gserviceaccount.com"
  + project   = "cloud-exercise-03-manassamant"
  + role      = "roles/run.invoker"
}
```

Plan: 1 to add, 0 to change, 0 to destroy.

Do you want to perform these actions?

Terraform will perform the actions described above.
Only 'yes' will be accepted to approve.

Enter a value: yes

```
module.iam.google_project_iam_member.cr_invoker: Creating...
module.iam.google_project_iam_member.cr_invoker: Creation complete after 9s [id=cloud-exercise-03-manassamant/roles/run.invoker/serviceAccount:cloud-run-sa@cloud-exercise-03-manassamant.iam.gserviceaccount.com]
```

Apply complete! Resources: 1 added, 0 changed, 0 destroyed.

Outputs:

```
cloud_function_service_account = "cloud-function-sa@cloud-exercise-03-manassamant.iam.gserviceaccount.com"
cloud_run_service_account       = "cloud-run-sa@cloud-exercise-03-manassamant.iam.gserviceaccount.com"
vpc_connector_id                = "projects/cloud-exercise-03-manassamant/locations/us-central1/connectors/exercise5-vpc-connector"
vpc_name                        = "exercise5-vpc"
```

C:\Users\Manas\Documents\exercise5>

```
C:\Users\Manas\Documents\exercise5>terraform apply
module.compute.data.archive_file.function_zip: Reading...
module.compute.data.archive_file.function_zip: Read complete after 0s [id=1c5dc8ea82dc7aff61145f05563ac960b749921b]
google_project_service.required_apis["cloudbuild.googleapis.com"]: Refreshing state... [id=cloud-exercise-03-manassamant/cloudbuild.googleapis.com]
module.storage.google_storage_bucket.terraform_state: Refreshing state... [id=cloud-exercise-03-manassamant-tf-state]
google_project_service.required_apis["compute.googleapis.com"]: Refreshing state... [id=cloud-exercise-03-manassamant/compute.googleapis.com]
module.iam.google_service_account.cloud_function_sa: Refreshing state... [id=projects/cloud-exercise-03-manassamant/serviceAccounts/cloud-function-sa@cloud-exercise-03-manassamant.iam.gserviceaccount.com]
google_project_service.required_apis["artifactregistry.googleapis.com"]: Refreshing state... [id=cloud-exercise-03-manassamant/artifactregistry.googleapis.com]
google_project_service.required_apis["cloudfunctions.googleapis.com"]: Refreshing state... [id=cloud-exercise-03-manassamant/cloudfunctions.googleapis.com]
module.iam.google_service_account.cloud_run_sa: Refreshing state... [id=projects/cloud-exercise-03-manassamant/serviceAccounts/cloud-run-sa@cloud-exercise-03-manassamant.iam.gserviceaccount.com]
google_project_service.required_apis["vpcaccess.googleapis.com"]: Refreshing state... [id=cloud-exercise-03-manassamant/vpcaccess.googleapis.com]
google_project_service.required_apis["run.googleapis.com"]: Refreshing state... [id=cloud-exercise-03-manassamant/run.googleapis.com]
google_project_service.required_apis["firestore.googleapis.com"]: Refreshing state... [id=cloud-exercise-03-manassamant/firestore.googleapis.com]
module.storage.google_storage_bucket.frontend: Refreshing state... [id=cloud-exercise-03-manassamant-frontend]
module.compute.google_storage_bucket.function_source: Refreshing state... [id=cloud-exercise-03-manassamant-function-source]
module.networking.google_compute_network.vpc: Refreshing state... [id=projects/cloud-exercise-03-manassamant/global/networks/exercise5-vpc]
module.iam.google_project_iam_member.cf_storage_access: Refreshing state... [id=cloud-exercise-03-manassamant/roles/storage.objectViewer/serviceAccount:cloud-function-sa@cloud-exercise-03-manassamant.iam.gserviceaccount.com]
module.iam.google_project_iam_member.cf_build_editor: Refreshing state... [id=cloud-exercise-03-manassamant/roles/cloudbuild.builds.builder/serviceAccount:cloud-function-sa@cloud-exercise-03-manassamant.iam.gserviceaccount.com]
module.iam.google_project_iam_member.cf_service_agent: Refreshing state... [id=cloud-exercise-03-manassamant/roles/cloudfunctions.serviceAgent/serviceAccount:cloud-function-sa@cloud-exercise-03-manassamant.iam.gserviceaccount.com]
module.iam.google_project_iam_member.cf_firestore_access: Refreshing state... [id=cloud-exercise-03-manassamant/roles/datastore.user/serviceAccount:cloud-function-sa@cloud-exercise-03-manassamant.iam.gserviceaccount.com]
module.iam.google_project_iam_member.cr_invoker: Refreshing state... [id=cloud-exercise-03-manassamant/roles/run.invoker/serviceAccount:cloud-run-sa@cloud-exercise-03-manassamant.iam.gserviceaccount.com]
module.storage.google_storage_bucket.iam_member.frontend_public_read: Refreshing state... [id=b/cloud-exercise-03-manassamant-frontend/roles/storage.objectViewer/allUsers]
module.storage.google_storage_bucket.object.frontend_files["index.html"]: Refreshing state... [id=cloud-exercise-03-manassamant-frontend-index.html]
module.networking.google_compute_firewall.allow_internal: Refreshing state... [id=projects/cloud-exercise-03-manassamant/global/firewalls/exercise5-vpc-allow-internal]
module.networking.google_compute_subnetwork.subnet: Refreshing state... [id=projects/cloud-exercise-03-manassamant/regions/us-central1/subnetworks/exercise5-vpc-subnet]
module.networking.google_vpc_access_connector.connector: Refreshing state... [id=projects/cloud-exercise-03-manassamant/locations/us-central1/connectors/exercise5-vpc-connector]
```

```
module.iam.google_project_iam_member.cf_firestore_access: Refreshing state... [id=cloud-exercise-03-manassamant/roles/datastore.user/serviceAccount:cloud-f
unction-sa@cloud-exercise-03-manassamant.iam.gserviceaccount.com]
module.iam.google_project_iam_member.cr_invoker: Refreshing state... [id=cloud-exercise-03-manassamant/roles/run.invoker/serviceAccount:cloud-run-sa@cloud-
exercise-03-manassamant.iam.gserviceaccount.com]
module.storage.google_storage_bucket_iam_member.frontend_public_read: Refreshing state... [id=b/cloud-exercise-03-manassamant-frontend/roles/storage.object
Viewer/allUsers]
module.storage.google_storage_bucket_object.frontend_files["index.html"]: Refreshing state... [id=cloud-exercise-03-manassamant-frontend-index.html]
module.networking.google_compute_firewall.allow_internal: Refreshing state... [id=projects/cloud-exercise-03-manassamant/global/firewalls/exercise5-vpc-all
ow-internal]
module.networking.google_compute_subnetwork.subnet: Refreshing state... [id=projects/cloud-exercise-03-manassamant/regions/us-central1/subnetworks/exercise
5-vpc-subnet]
module.networking.google_vpc_access_connector.connector: Refreshing state... [id=projects/cloud-exercise-03-manassamant/locations/us-central1/connectors/ex
ercise5-vpc-connector]
module.compute.google_storage_bucket_object.function_zip: Refreshing state... [id=cloud-exercise-03-manassamant-function-source-function.zip]
module.compute.google_cloudfunctions2_function.backend: Refreshing state... [id=projects/cloud-exercise-03-manassamant/locations/us-central1/functions/exer
cise5-backend]
google_cloud_run_v2_service_iam_member.function_public_invoker: Refreshing state... [id=projects/cloud-exercise-03-manassamant/locations/us-central1/servic
es/exercise5-backend/roles/run.invoker/allUsers]
google_cloud_run_v2_service_iam_member.backend_public: Refreshing state... [id=projects/cloud-exercise-03-manassamant/locations/us-central1/services/exerci
se5-backend/roles/run.invoker/allUsers]
```

Terraform used the selected providers to generate the following execution plan. Resource actions are indicated with the following symbols:
~ update in-place

Terraform will perform the following actions:

```
# module.compute.google_cloudfunctions2_function.backend will be updated in-place
~ resource "google_cloudfunctions2_function" "backend" {
  id      = "projects/cloud-exercise-03-manassamant/locations/us-central1/functions/exercise5-backend"
  name    = "exercise5-backend"
  # (11 unchanged attributes hidden)

  ~ service_config {
    - vpc_connector_egress_settings = "PRIVATE_RANGES_ONLY" -> null
    # (14 unchanged attributes hidden)
  }

  # (1 unchanged block hidden)
}
```

Plan: 0 to add, 1 to change, 0 to destroy.



```
    # (1 unchanged block hidden)
  }

Plan: 0 to add, 1 to change, 0 to destroy.

Do you want to perform these actions?
  Terraform will perform the actions described above.
  Only 'yes' will be accepted to approve.

  Enter a value: yes

module.compute.google_cloudfunctions2_function.backend: Modifying... [id=projects/cloud-exercise-03-manassamant/locations/us-central1/functions/exercise5-backend]
module.compute.google_cloudfunctions2_function.backend: Still modifying... [id=projects/cloud-exercise-03-manassamant...s-central1/functions/exercise5-backend, 00m10s elapsed]
module.compute.google_cloudfunctions2_function.backend: Still modifying... [id=projects/cloud-exercise-03-manassamant...s-central1/functions/exercise5-backend, 00m20s elapsed]
module.compute.google_cloudfunctions2_function.backend: Still modifying... [id=projects/cloud-exercise-03-manassamant...s-central1/functions/exercise5-backend, 00m30s elapsed]
module.compute.google_cloudfunctions2_function.backend: Still modifying... [id=projects/cloud-exercise-03-manassamant...s-central1/functions/exercise5-backend, 00m40s elapsed]
module.compute.google_cloudfunctions2_function.backend: Still modifying... [id=projects/cloud-exercise-03-manassamant...s-central1/functions/exercise5-backend, 00m50s elapsed]
module.compute.google_cloudfunctions2_function.backend: Modifications complete after 58s [id=projects/cloud-exercise-03-manassamant/locations/us-central1/functions/exercise5-backend]

Apply complete! Resources: 0 added, 1 changed, 0 destroyed.

Outputs:

cloud_function_service_account = "cloud-function-sa@cloud-exercise-03-manassamant.iam.gserviceaccount.com"
cloud_function_url = "https://exercise5-backend-l7oasm6fwq-uc.a.run.app"
cloud_run_service_account = "cloud-run-sa@cloud-exercise-03-manassamant.iam.gserviceaccount.com"
frontend_bucket = "cloud-exercise-03-manassamant-frontend"
frontend_url = "https://storage.googleapis.com/cloud-exercise-03-manassamant-frontend/index.html"
terraform_state_bucket = "cloud-exercise-03-manassamant-tf-state"
vpc_connector_id = "projects/cloud-exercise-03-manassamant/locations/us-central1/connectors/exercise5-vpc-connector"
vpc_name = "exercise5-vpc"

C:\Users\Manas\Documents\exercise5>
```

Terraform Show

```
Command Prompt
C:\Users\Manas\Documents\exercise5>terraform output
cloud_function_service_account = "cloud-function-sa@cloud-exercise-03-manassamant.iam.gserviceaccount.com"
cloud_function_url = "https://exercise5-backend-l7oasw6fwq-uc.a.run.app"
cloud_run_service_account = "cloud-run-sa@cloud-exercise-03-manassamant.iam.gserviceaccount.com"
frontend_bucket = "cloud-exercise-03-manassamant-frontend"
frontend_url = "https://storage.googleapis.com/cloud-exercise-03-manassamant-frontend/index.html"
terraform_state_bucket = "cloud-exercise-03-manassamant-tf-state"
vpc_connector_id = "projects/cloud-exercise-03-manassamant/locations/us-central1/connectors/exercise5-vpc-connector"
vpc_name = "exercise5-vpc"

C:\Users\Manas\Documents\exercise5>notepad C:\Users\Manas\Documents\exercise5\providers.tf

C:\Users\Manas\Documents\exercise5>notepad C:\Users\Manas\Documents\exercise5\providers.tf

C:\Users\Manas\Documents\exercise5>terraform show
# google_cloud_run_v2_service_iam_member.backend_public:
resource "google_cloud_run_v2_service_iam_member" "backend_public" {
  etag      = "BwZIJ2I6CYQ="
  id        = "projects/cloud-exercise-03-manassamant/locations/us-central1/services/exercise5-backend/roles/run.invoker/allUsers"
  location  = "us-central1"
  member    = "allUsers"
  name      = "projects/cloud-exercise-03-manassamant/locations/us-central1/services/exercise5-backend"
  project   = "cloud-exercise-03-manassamant"
  role      = "roles/run.invoker"
}

# google_cloud_run_v2_service_iam_member.function_public_invoker:
resource "google_cloud_run_v2_service_iam_member" "function_public_invoker" {
  etag      = "BwZIJ2I6CYQ="
  id        = "projects/cloud-exercise-03-manassamant/locations/us-central1/services/exercise5-backend/roles/run.invoker/allUsers"
  location  = "us-central1"
  member    = "allUsers"
  name      = "projects/cloud-exercise-03-manassamant/locations/us-central1/services/exercise5-backend"
  project   = "cloud-exercise-03-manassamant"
  role      = "roles/run.invoker"
}

# google_project_service.required_apis["artifactregistry.googleapis.com"]:
resource "google_project_service" "required_apis" {
  disable_on_destroy = false
  id                  = "cloud-exercise-03-manassamant/artifactregistry.googleapis.com"
}
```

```
Command Prompt
member = "serviceAccount:cloud-function-sa@cloud-exercise-03-manassamant.iam.gserviceaccount.com"
project = "cloud-exercise-03-manassamant"
role = "roles/datastore.user"
}

# module.iam.google_project_iam_member.cf_service_agent:
resource "google_project_iam_member" "cf_service_agent" {
  etag = "BwZIJkWRudk="
  id = "cloud-exercise-03-manassamant/roles/cloudfunctions.serviceAgent/serviceAccount:cloud-function-sa@cloud-exercise-03-manassamant.iam.gserviceaccount.com"
  member = "serviceAccount:cloud-function-sa@cloud-exercise-03-manassamant.iam.gserviceaccount.com"
  project = "cloud-exercise-03-manassamant"
  role = "roles/cloudfunctions.serviceAgent"
}

# module.iam.google_project_iam_member.cf_storage_access:
resource "google_project_iam_member" "cf_storage_access" {
  etag = "BwZIJkWRudk="
  id = "cloud-exercise-03-manassamant/roles/storage.objectViewer/serviceAccount:cloud-function-sa@cloud-exercise-03-manassamant.iam.gserviceaccount.com"
  member = "serviceAccount:cloud-function-sa@cloud-exercise-03-manassamant.iam.gserviceaccount.com"
  project = "cloud-exercise-03-manassamant"
  role = "roles/storage.objectViewer"
}

# module.iam.google_project_iam_member.cr_invoker:
resource "google_project_iam_member" "cr_invoker" {
  etag = "BwZIJkWRudk="
  id = "cloud-exercise-03-manassamant/roles/run.invoker/serviceAccount:cloud-run-sa@cloud-exercise-03-manassamant.iam.gserviceaccount.com"
  member = "serviceAccount:cloud-run-sa@cloud-exercise-03-manassamant.iam.gserviceaccount.com"
  project = "cloud-exercise-03-manassamant"
  role = "roles/run.invoker"
}

# module.iam.google_service_account.cloud_function_sa:
resource "google_service_account" "cloud_function_sa" {
  account_id = "cloud-function-sa"
  description = null
  disabled = false
  display_name = "Cloud Function Service Account"
  email = "cloud-function-sa@cloud-exercise-03-manassamant.iam.gserviceaccount.com"
}
```

Terraform Output

```
C:\Users\Manas\Documents\exercise5>cd C:\Users\Manas\Documents\exercise5

C:\Users\Manas\Documents\exercise5>terraform output
cloud_function_service_account = "cloud-function-sa@cloud-exercise-03-manassamant.iam.gserviceaccount.com"
cloud_function_url = "https://exercise5-backend-l7oasw6fwq-uc.a.run.app"
cloud_run_service_account = "cloud-run-sa@cloud-exercise-03-manassamant.iam.gserviceaccount.com"
frontend_bucket = "cloud-exercise-03-manassamant-frontend"
frontend_url = "https://storage.googleapis.com/cloud-exercise-03-manassamant-frontend/index.html"
terraform_state_bucket = "cloud-exercise-03-manassamant-tf-state"
vpc_connector_id = "projects/cloud-exercise-03-manassamant/locations/us-central1/connectors/exercise5-vpc-connector"
vpc_name = "exercise5-vpc"

C:\Users\Manas\Documents\exercise5>
```



Curl Command Output

Apply complete! Resources: 1 added, 1 changed, 1 destroyed.

Outputs:

```
cloud_function_service_account = "cloud-function-sa@cloud-exercise-03-manassamant.iam.gserviceaccount.com"
cloud_function_url = "https://exercise5-backend-l7oasw6fwq-uc.a.run.app"
cloud_run_service_account = "cloud-run-sa@cloud-exercise-03-manassamant.iam.gserviceaccount.com"
frontend_bucket = "cloud-exercise-03-manassamant-frontend"
frontend_url = "https://storage.googleapis.com/cloud-exercise-03-manassamant-frontend/index.html"
terraform_state_bucket = "cloud-exercise-03-manassamant-tf-state"
vpc_connector_id = "projects/cloud-exercise-03-manassamant/locations/us-central1/connectors/exercise5-vpc-connector"
vpc_name = "exercise5-vpc"
```

```
C:\Users\Manas\Documents\exercise5>gcloud run services add-iam-policy-binding exercise5-backend ^
```

```
More? --region=us-central1 ^
```

```
More? --member="allUsers" ^
```

```
More? --role="roles/run.invoker"
```

```
Updated IAM policy for service [exercise5-backend].
```

```
bindings:
```

```
- members:
```

```
  - allUsers
```

```
    role: roles/run.invoker
```

```
etag: BwZIJ2I6CYQ=
```

```
version: 1
```

```
C:\Users\Manas\Documents\exercise5>curl "https://exercise5-backend-l7oasw6fwq-uc.a.run.app?name=Manas"
```

```
{ "error": "404 The database (default) does not exist for project cloud-exercise-03-manassamant Please visit https://console.cloud.google.com/datastore/setu  
p?project=cloud-exercise-03-manassamant to add a Cloud Datastore or Cloud Firestore database. " }
```

```
C:\Users\Manas\Documents\exercise5>curl "https://exercise5-backend-l7oasw6fwq-uc.a.run.app?name=Manas"
```

```
{ "message": "Hello, Manas!", "count": 1 }
```

```
C:\Users\Manas\Documents\exercise5>curl "https://exercise5-backend-l7oasw6fwq-uc.a.run.app?name=Manas"
```

```
{ "message": "Hello, Manas!", "count": 2 }
```

```
C:\Users\Manas\Documents\exercise5>
```



Greetings SS

console.cloud.google.com/firestore/databases/-/default-/data/panel/greetings/Manas?project=cloud-exercise-03-manassamant

Free trial status: €256.71 credit and 42 days remaining. Activate your full account to get unlimited access to all of Google Cloud – use any remaining credits, then pay only for what you use. Dismiss Activate

Google Cloud cloud-exercise-03-ManasSamant firestore

Firestore

Database

Panel view Query builder

Firestore studio / > greetings > Manas

(default)	greetings	Manas
+ Start collection	+ Add document	+ Start collection
greetings >	Manas >	+ Add field count: 2

Precip coming In about 1.5 hours

13:36 12-01-2026

GCP Console - Cloud Function details page

The screenshot shows the Google Cloud console interface. On the left, the 'Services' page is active. The main content area shows a table of services. The first service is 'exercise5-backend', which is a 'Function' deployment type, located in the 'us-central1' region, with 'Public access' authentication, 'All' ingress, and was last deployed '10 hours ago' by 'Cloud Run functions'.

Name	Deployment type	Req/sec	Region	Authentication	Ingress	Last deployed	Deployed by
exercise5-backend	Function	0	us-central1	Public access	All	10 hours ago	Cloud Run functions

GCP Console - Cloud Run service details page

The screenshot displays the Google Cloud Console interface for a Cloud Storage bucket. The browser address bar shows the URL: `console.cloud.google.com/storage/browser/cloud-exercise-03-manassamant-frontend?tab=objects&forceOnBucketsSortingFiltering=true&project=cloud-exerci...`. The page title is "cloud-exercise-03-ManasSamant" with a search bar and a "buck" filter. The left sidebar contains navigation links for Cloud Storage, Storage Intelligence, Marketplace, and Release notes. The main content area shows the "Bucket details" for "cloud-exercise-03-manassamant-frontend". The bucket's location is "us-central1 (Iowa)", storage class is "Standard", and public access is "Access granted to public principals". The protection setting is "Soft delete". Below the details, the "Objects" tab is active, showing a folder browser and a table of objects. The table has columns for Name, Size, and Type, and contains one object named "index.html" with a size of 331 B and type "text/html; charset=...". The right sidebar features a "Get started with Cloud Storage" section with links to "Getting bucket information", "Uploading objects", "Downloading objects", "Use cases for Cloud Storage", and "Terraform samples". The bottom status bar shows the system clock at 03:56 on 12-01-2026.

Free trial status: €256.71 credit and 42 days remaining. Activate your full account to get unlimited access to all of Google Cloud – use any remaining credits, then pay only for what you use.

Google Cloud cloud-exercise-03-ManasSamant buck Search

Cloud Storage Bucket details Edit bucket Go to path Refresh

cloud-exercise-03-manassamant-frontend

Location: us-central1 (Iowa) Storage class: Standard Public access: Access granted to public principals Protection: Soft delete

Objects Configuration Permissions Protection Lifecycle Observability New Inventory Reports

Folder browser cloud-exercise-03-manassamant-frontend

Buckets > cloud-exercise-03-manassamant-frontend

Create folder Upload Transfer data Other services

Filter by name prefix only Filter Show Live objects only

Name	Size	Type
index.html	331 B	text/html; charset=...

Get started with Cloud Storage

- Getting bucket information Help document Get information on the size and metadata of your Cloud Storage buckets.
- Uploading objects Help document Upload the objects containing your data to your Cloud Storage buckets.
- Downloading objects Help document Download the objects from your Cloud Storage buckets.
- Use cases for Cloud Storage Help document Explore use cases, best practices, and industry solutions.
- Terraform samples Help document See examples of using Terraform to create Cloud Storage resources.

03:56 12-01-2026

Cloud Logging showing function invocation logs with timestamps

console.cloud.google.com/run/detail/us-central1/exercise5-backend/observability/logs?project=cloud-exercise-03-manassamant

Free trial status: €255.71 credit and 42 days remaining. Activate your full account to get unlimited access to all of Google Cloud – use any remaining credits, then pay only for what you use.

Google Cloud cloud-exercise-03-ManasSamant cloud functions Search

Cloud Run Service details Edit and deploy new revision Connect to repo Test Learn Refresh

exercise5-backend Region: us-central1 URL: https://us-central1-cloud-exercise-03-manassamant.cloudfunctions.net/exercise5-backend Scaling: Auto (min: 0, r

Overview Services Jobs Worker pools Domain mappings

Observability Revisions Source Triggers Networking Security YAML

Metrics Logs Severity: Default Filter Search all fields and values

Severity	Timestamp	Summary
> i	2026-01-12 03:10:12.428 CET	Cloud Run CreateInternalService exercise5-backend {@type: type.googleapis.com/google.cloud.audit.AuditLog, metho...
> i	2026-01-12 03:12:36.457 CET	GET 200 498 B 845 ms curl/8.16.0 https://exercise5-backend-l7oasw6fwq-uc.a.run.app/?name=Manas
> i	2026-01-12 03:19:19.121 CET	GET 200 276 B 597 ms curl/8.16.0 https://exercise5-backend-l7oasw6fwq-uc.a.run.app/?name=Manas
> i	2026-01-12 03:19:32.928 CET	GET 200 276 B 486 ms curl/8.16.0 https://exercise5-backend-l7oasw6fwq-uc.a.run.app/?name=Manas
> i	2026-01-12 03:29:36.121 CET	Cloud Run ReplaceInternalService exercise5-backend service-1085206388614@pcf-admin-robot... {@type: type.googleapi...
> i	2026-01-12 03:30:15.830 CET	Cloud Run ReplaceInternalService exercise5-backend service-1085206388614@pcf-admin-robot... {@type: type.googleapi...
> i	2026-01-12 03:30:24.229 CET	Starting new instance. Reason: DEPLOYMENT_ROLLOUT - Instance started due to traffic shifting between revisions due to...
> i	2026-01-12 03:30:26.813 CET	Default STARTUP TCP probe succeeded after 1 attempt for container "worker" on port 8080.
> i	2026-01-12 03:30:26.924 CET	Cloud Run ReplaceInternalService exercise5-backend-80002-tor {@type: type.googleapis.com/google.cloud.audit.Audi...
> i	2026-01-12 03:30:28.576 CET	Cloud Run ReplaceInternalService exercise5-backend {@type: type.googleapis.com/google.cloud.audit.AuditLog, metho...

No newer entries found matching current filter.

2°C Cloudy Search 13:50 12-01-2026

The screenshot displays the Google Cloud console interface for the 'exercise5-backend' service. The 'Metrics' tab is active, showing four empty charts: 'Request count', 'Request latencies', 'End-to-end request latency', and 'Latency breakdown'. The left sidebar contains navigation links for Overview, Services, Jobs, Worker pools, and Domain mappings. The top navigation bar includes the project name 'cloud-exercise-03-ManasSamant' and the service name 'cloud functions'. The bottom status bar shows the system clock as 13:49 on 12-01-2026.

Terraform state list

Terraform has been successfully initialized!

You may now begin working with Terraform. Try running "terraform plan" to see any changes that are required for your infrastructure. All Terraform commands should now work.

If you ever set or change modules or backend configuration for Terraform, rerun this command to reinitialize your working directory. If you forget, other commands will detect it and remind you to do so if necessary.

C:\Users\Manas\Documents\exercise5>

```
C:\Users\Manas\Documents\exercise5>terraform state list
google_project_service.required_api["artifactregistry.googleapis.com"]
google_project_service.required_api["cloudbuild.googleapis.com"]
google_project_service.required_api["cloudfunctions.googleapis.com"]
google_project_service.required_api["compute.googleapis.com"]
google_project_service.required_api["firestore.googleapis.com"]
google_project_service.required_api["run.googleapis.com"]
google_project_service.required_api["vpcaccess.googleapis.com"]
module.compute.data.archive_file.function_zip
module.compute.google_cloudfunctions2_function.backend
module.compute.google_storage_bucket.function_source
module.compute.google_storage_bucket_object.function_zip
module.iam.google_project_iam_member.cf_build_editor
module.iam.google_project_iam_member.cf_firestore_access
module.iam.google_project_iam_member.cf_service_agent
module.iam.google_project_iam_member.cf_storage_access
module.iam.google_project_iam_member.cr_invoker
module.iam.google_service_account.cloud_function_sa
module.iam.google_service_account.cloud_run_sa
module.networking.google_compute_firewall.allow_internal
module.networking.google_compute_network.vpc
module.networking.google_compute_subnetwork.subnet
module.networking.google_vpc_access_connector.connector
module.storage.google_storage_bucket.frontend
module.storage.google_storage_bucket.terraform_state
module.storage.google_storage_bucket_iam_member.frontend_public_read
module.storage.google_storage_bucket_object.frontend_files["index.html"]
```

C:\Users\Manas\Documents\exercise5>



Free trial status: €256.71 credit and 43 days remaining. Activate your full account to get unlimited access to all of Google Cloud – use any remaining credits, then pay only for what you use. [Dismiss](#) [Activate](#)

Google Cloud cloud-exercise-03-ManasSamant firestore [Search](#)

Firestore All databases

Firestore is an enterprise-grade, serverless document database service – now with MongoDB compatibility

Easily store and read semi-structured JSON data on a differentiated Firestore database service with virtually unlimited scalability, industry-leading availability with up to 99.999% SLA and single-digit milliseconds read latency using flexible Firestore, Datastore or MongoDB-compatible interfaces.

[Create a Firestore database](#) [View fleet health](#)

[Show deleted](#)

Database ID	Edition	Configuration	Location	Creation time (UTC+1)	Scheduled backups	Encryption type	Encryption key	Actions
(default)	Standard	Firestore native	us-central1 (Iowa)	12 Jan 2026, 03:16:17	Disabled Edit settings	Google-managed		

Firewall Policies

The screenshot displays the Google Cloud console interface for managing Firewall Policies. The main content area shows a table of network firewall policies. The table has columns for Name, Type, Targets, Filters, Protocols/ports, Action, and Priority. The policies listed include 'cloud-exercise-03-manassamant-dev-allow-internal', 'default-allow-icmp', 'default-allow-internal', 'default-allow-rdp', 'default-allow-ssh', and 'exercise5-vpc-allow-internal'. Each policy is of type 'Ingress' and applies to all targets. The actions are 'Allow' for all policies. The priorities are 100 for the first and last policies, and 6553 for the others.

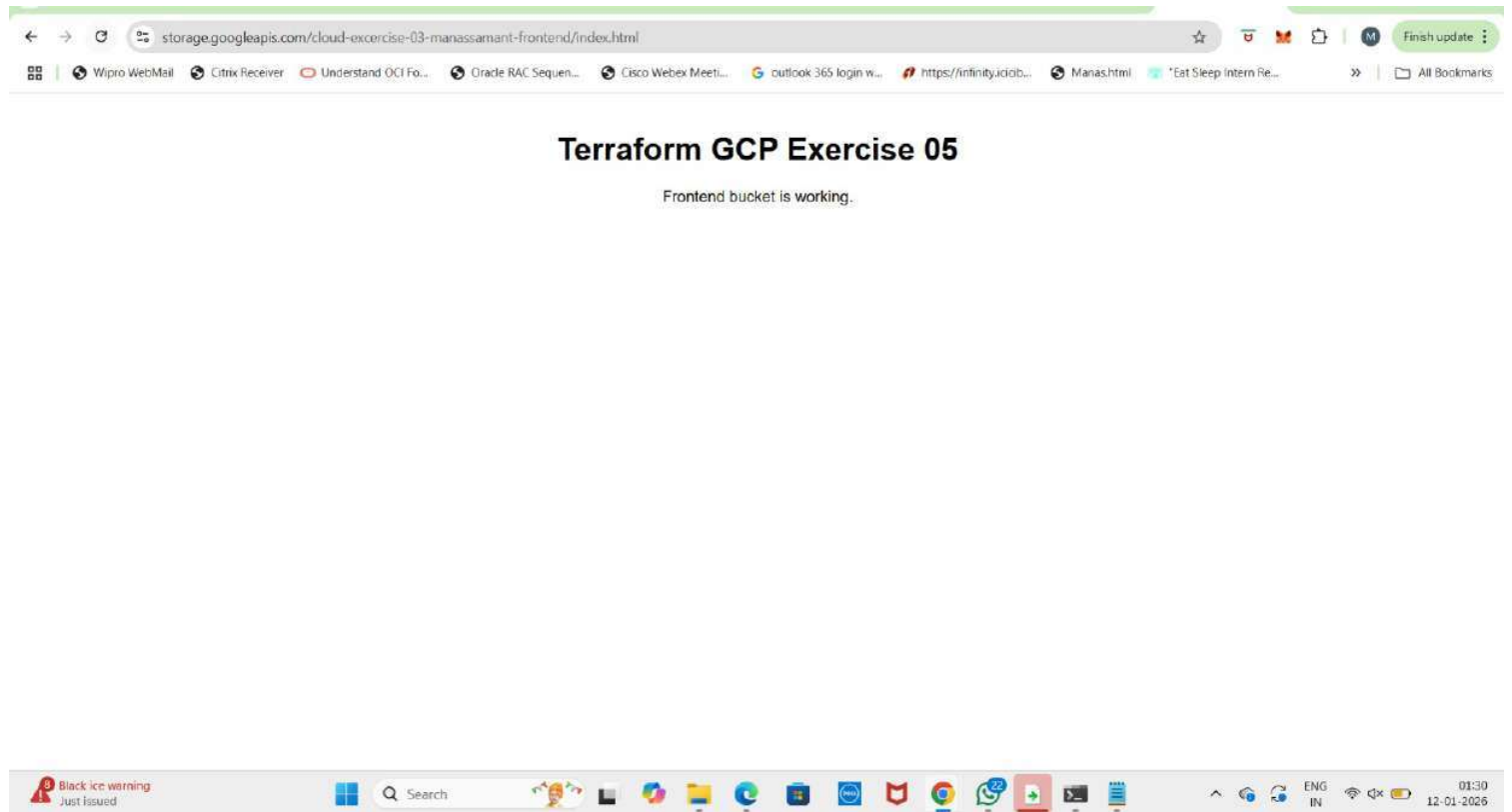
Name	Type	Targets	Filters	Protocols/ports	Action	Priority
cloud-exercise-03-manassamant-dev-allow-internal	Ingress	Apply to all	IP ranges:	tcp:0-65535	Allow	100
default-allow-icmp	Ingress	Apply to all	IP ranges:	icmp	Allow	6553
default-allow-internal	Ingress	Apply to all	IP ranges:	tcp:0-65535 udp:0-65535 icmp	Allow	6553
default-allow-rdp	Ingress	Apply to all	IP ranges:	tcp:3389	Allow	6553
default-allow-ssh	Ingress	Apply to all	IP ranges:	tcp:22	Allow	6553
exercise5-vpc-allow-internal	Ingress	Apply to all	IP ranges:	tcp:0-65535 udp:0-65535	Allow	100

Network firewall policies

Recommended for you

- [Cloud Next Generation Firewall overview](#)
Help document
Choose a fully distributed firewall service with advanced protection capabilities to safeguard your Google Cloud workloads from internal and external attacks.
- [Types of firewall policies](#)
Help document
Group firewall rules into policy objects that apply to an entire organization, a VPC network, or a region.
- [Create a global network firewall policy](#)
Help document
Create a global network firewall policy and add rules to it.
- [Configure intrusion prevention service](#)
Help document
Set up intrusion prevention service in your network to protect your workload traffic from threats such as intrusions.

Frontend URL



gcloud list

```
t{"message": "Hello, Manas!", "count": 1}
C:\Users\Manas\Documents\exercise5>curl "https://exercise5-backend-l7oasw6fwq-uc.a.run.app?name=Manas"
{"message": "Hello, Manas!", "count": 2}
C:\Users\Manas\Documents\exercise5>gcloud auth list
Credentialed Accounts

ACTIVE: *
ACCOUNT: manassamant4@gmail.com

To set the active account, run:
$ gcloud config set account 'ACCOUNT'

C:\Users\Manas\Documents\exercise5>gcloud config list project
[core]
project = cloud-exercise-03-manassamant

Your active configuration is: [default]

C:\Users\Manas\Documents\exercise5>
```



gcs_backend_compute

```
vpc_name = "exercise5-vpc"

C:\Users\Manas\Documents\exercise5>cd C:\Users\Manas\Documents\exercise5

C:\Users\Manas\Documents\exercise5>terraform output terraform_state_bucket
"cloud-exercise-03-manassamant-tf-state"

C:\Users\Manas\Documents\exercise5>notepad providers.tf

C:\Users\Manas\Documents\exercise5>terraform init -migrate-state
Initializing the backend...
Acquiring state lock. This may take a few moments...
Do you want to copy existing state to the new backend?
  Pre-existing state was found while migrating the previous "local" backend to the
  newly configured "gcs" backend. No existing state was found in the newly
  configured "gcs" backend. Do you want to copy this state to the new "gcs"
  backend? Enter "yes" to copy and "no" to start with an empty state.

Enter a value: yes

Successfully configured the backend "gcs"! Terraform will automatically
use this backend unless the backend configuration changes.
Initializing modules...
Initializing provider plugins...
- Reusing previous version of hashicorp/google from the dependency lock file
- Reusing previous version of hashicorp/archive from the dependency lock file
- Using previously-installed hashicorp/archive v2.7.1
- Using previously-installed hashicorp/google v5.45.2

Terraform has been successfully initialized!

You may now begin working with Terraform. Try running "terraform plan" to see
any changes that are required for your infrastructure. All Terraform commands
should now work.

If you ever set or change modules or backend configuration for Terraform,
rerun this command to reinitialize your working directory. If you forget, other
commands will detect it and remind you to do so if necessary.

C:\Users\Manas\Documents\exercise5>
```

```

Command Prompt
google_project_service.required_apis["firestore.googleapis.com"]: Creation complete after 4s [id=cloud-exercise-03-manassamant/firestore.googleapis.com]
google_project_service.required_apis["compute.googleapis.com"]: Creation complete after 4s [id=cloud-exercise-03-manassamant/compute.googleapis.com]
google_project_service.required_apis["cloudbuild.googleapis.com"]: Creation complete after 4s [id=cloud-exercise-03-manassamant/cloudbuild.googleapis.com]
google_project_service.required_apis["cloudfunctions.googleapis.com"]: Creation complete after 4s [id=cloud-exercise-03-manassamant/cloudfunctions.googleapis.com]
google_project_service.required_apis["vpcaccess.googleapis.com"]: Creation complete after 4s [id=cloud-exercise-03-manassamant/vpcaccess.googleapis.com]
google_project_service.required_apis["artifactregistry.googleapis.com"]: Creation complete after 4s [id=cloud-exercise-03-manassamant/artifactregistry.googleapis.com]
module.networking.google_compute_network.vpc: Still creating... [00m10s elapsed]
module.networking.google_compute_network.vpc: Creation complete after 12s [id=projects/cloud-exercise-03-manassamant/global/networks/exercise5-vpc]
module.networking.google_vpc_access_connector.connector: Creating...
module.networking.google_compute_subnetwork.subnet: Creating...
module.networking.google_compute_firewall.allow_internal: Creating...
module.networking.google_vpc_access_connector.connector: Still creating... [00m10s elapsed]
module.networking.google_compute_subnetwork.subnet: Still creating... [00m10s elapsed]
module.networking.google_compute_firewall.allow_internal: Still creating... [00m10s elapsed]
module.networking.google_compute_firewall.allow_internal: Creation complete after 12s [id=projects/cloud-exercise-03-manassamant/global/firewalls/exercise5-vpc-allow-internal]
module.networking.google_compute_subnetwork.subnet: Creation complete after 14s [id=projects/cloud-exercise-03-manassamant/regions/us-central1/subnetworks/exercise5-vpc-subnet]
module.networking.google_vpc_access_connector.connector: Still creating... [00m20s elapsed]
module.networking.google_vpc_access_connector.connector: Still creating... [00m30s elapsed]
module.networking.google_vpc_access_connector.connector: Still creating... [00m40s elapsed]
module.networking.google_vpc_access_connector.connector: Still creating... [00m50s elapsed]
module.networking.google_vpc_access_connector.connector: Still creating... [01m00s elapsed]
module.networking.google_vpc_access_connector.connector: Still creating... [01m10s elapsed]
module.networking.google_vpc_access_connector.connector: Still creating... [01m20s elapsed]
module.networking.google_vpc_access_connector.connector: Still creating... [01m30s elapsed]
module.networking.google_vpc_access_connector.connector: Still creating... [01m40s elapsed]
module.networking.google_vpc_access_connector.connector: Still creating... [01m50s elapsed]
module.networking.google_vpc_access_connector.connector: Creation complete after 1m58s [id=projects/cloud-exercise-03-manassamant/locations/us-central1/connectors/exercise5-vpc-connector]

Apply complete! Resources: 11 added, 0 changed, 0 destroyed.

Outputs:
vpc_connector_id = "projects/cloud-exercise-03-manassamant/locations/us-central1/connectors/exercise5-vpc-connector"
vpc_name = "exercise5-vpc"

C:\Users\Manas\Documents\exercise5>

```



```

+ timeout_seconds      = 60
+ uri                  = (known after apply)
+ vpc_connector         = "projects/cloud-exercise-03-manassamant/locations/us-central1/connectors/exercise5-vpc-connector"
}
}

Plan: 1 to add, 0 to change, 0 to destroy.

Changes to Outputs:
+ cloud_function_url    = (known after apply)

Do you want to perform these actions?
  Terraform will perform the actions described above.
  Only 'yes' will be accepted to approve.

  Enter a value: yes

module.compute.google_cloudfunctions2_function.backend: Creating...
module.compute.google_cloudfunctions2_function.backend: Still creating... [00m10s elapsed]
module.compute.google_cloudfunctions2_function.backend: Still creating... [00m20s elapsed]
module.compute.google_cloudfunctions2_function.backend: Still creating... [00m30s elapsed]
module.compute.google_cloudfunctions2_function.backend: Still creating... [00m40s elapsed]
module.compute.google_cloudfunctions2_function.backend: Still creating... [00m50s elapsed]
module.compute.google_cloudfunctions2_function.backend: Still creating... [01m00s elapsed]
module.compute.google_cloudfunctions2_function.backend: Creation complete after 1m9s [id=projects/cloud-exercise-03-manassamant/locations/us-central1/functions/exercise5-backend]

Apply complete! Resources: 1 added, 0 changed, 0 destroyed.

Outputs:

cloud_function_service_account = "cloud-function-sa@cloud-exercise-03-manassamant.iam.gserviceaccount.com"
cloud_function_url              = "https://exercise5-backend-l7oasw6fwq-uc.a.run.app"
cloud_run_service_account       = "cloud-run-sa@cloud-exercise-03-manassamant.iam.gserviceaccount.com"
frontend_bucket                 = "cloud-exercise-03-manassamant-frontend"
frontend_url                    = "https://storage.googleapis.com/cloud-exercise-03-manassamant-frontend/index.html"
terraform_state_bucket         = "cloud-exercise-03-manassamant-tf-state"
vpc_connector_id                = "projects/cloud-exercise-03-manassamant/locations/us-central1/connectors/exercise5-vpc-connector"
vpc_name                        = "exercise5-vpc"

C:\Users\Manas\Documents\exercise5>

```

Apply complete! Resources: 1 added, 1 changed, 1 destroyed.

Outputs:

```
cloud_function_service_account = "cloud-function-sa@cloud-exercise-03-manassamant.iam.gserviceaccount.com"
cloud_function_url = "https://exercise5-backend-l7oasw6fwq-uc.a.run.app"
cloud_run_service_account = "cloud-run-sa@cloud-exercise-03-manassamant.iam.gserviceaccount.com"
frontend_bucket = "cloud-exercise-03-manassamant-frontend"
frontend_url = "https://storage.googleapis.com/cloud-exercise-03-manassamant-frontend/index.html"
terraform_state_bucket = "cloud-exercise-03-manassamant-tf-state"
vpc_connector_id = "projects/cloud-exercise-03-manassamant/locations/us-central1/connectors/exercise5-vpc-connector"
vpc_name = "exercise5-vpc"
```

```
C:\Users\Manas\Documents\exercise5>gcloud run services add-iam-policy-binding exercise5-backend ^
```

```
More? --region=us-central1 ^
```

```
More? --member="allUsers" ^
```

```
More? --role="roles/run.invoker"
```

```
Updated IAM policy for service [exercise5-backend].
```

```
bindings:
```

```
- members:
  - allUsers
    role: roles/run.invoker
```

```
etag: BwZIJ2I6CYQ=
```

```
version: 1
```

```
C:\Users\Manas\Documents\exercise5>curl "https://exercise5-backend-l7oasw6fwq-uc.a.run.app?name=Manas"
```

```
{"error": "404 The database (default) does not exist for project cloud-exercise-03-manassamant Please visit https://console.cloud.google.com/datastore/setu  
p?project=cloud-exercise-03-manassamant to add a Cloud Datastore or Cloud Firestore database. "}
```

```
C:\Users\Manas\Documents\exercise5>curl "https://exercise5-backend-l7oasw6fwq-uc.a.run.app?name=Manas"
```

```
{"message": "Hello, Manas!", "count": 1}
```

```
C:\Users\Manas\Documents\exercise5>curl "https://exercise5-backend-l7oasw6fwq-uc.a.run.app?name=Manas"
```

```
{"message": "Hello, Manas!", "count": 2}
```

```
C:\Users\Manas\Documents\exercise5>
```

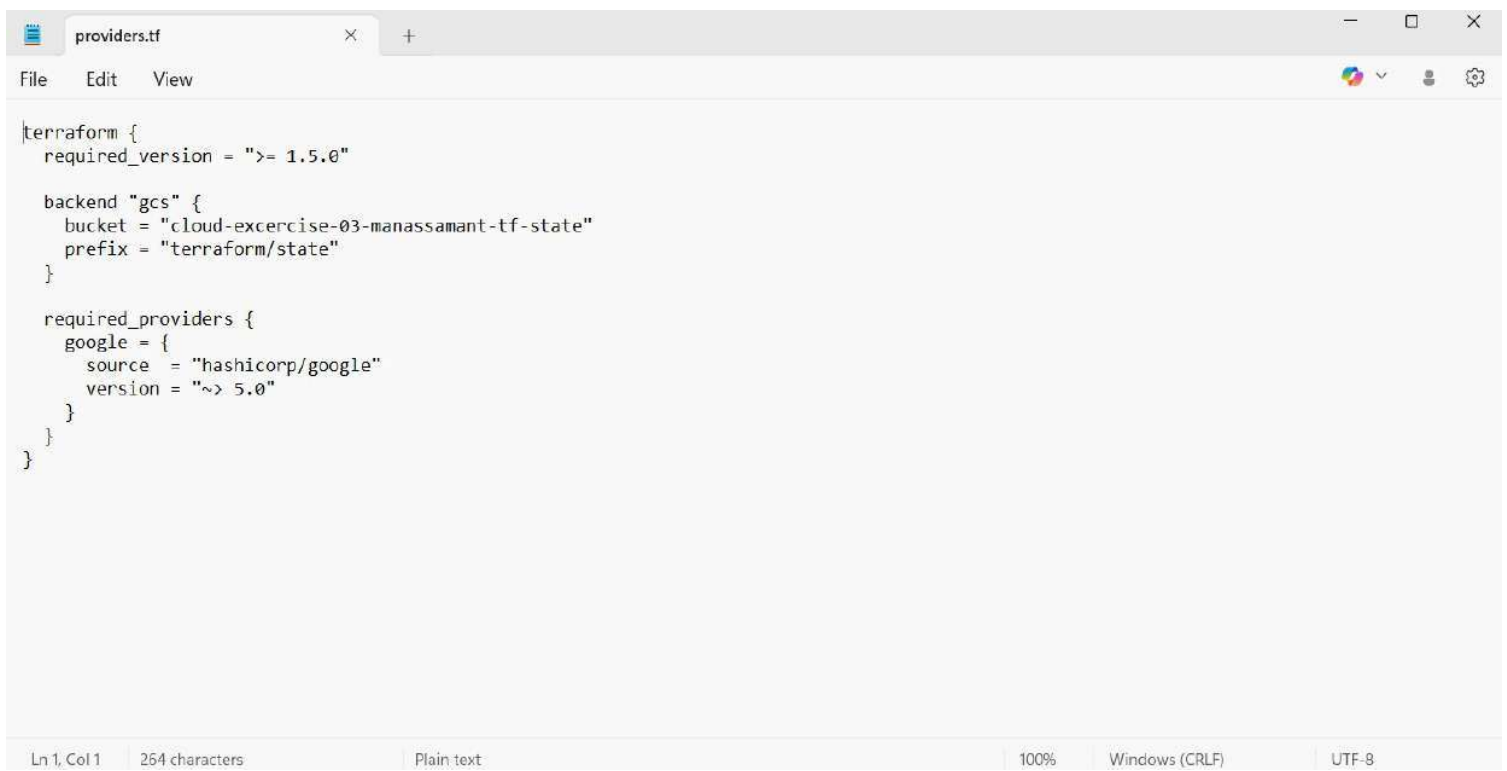


Black ice warning
Starts soon



03:19
12-01-2026

providers_backend_block_ss



The image shows a code editor window with a single tab titled "providers.tf". The editor has a menu bar with "File", "Edit", and "View". The main area contains the following Terraform configuration:

```
terraform {  
  required_version = ">= 1.5.0"  
  
  backend "gcs" {  
    bucket = "cloud-exercise-03-manassamant-tf-state"  
    prefix = "terraform/state"  
  }  
  
  required_providers {  
    google = {  
      source = "hashicorp/google"  
      version = "~> 5.0"  
    }  
  }  
}
```

The status bar at the bottom indicates the cursor is at "Ln 1, Col 1", the file contains "254 characters", the encoding is "Plain text", the zoom is "100%", the line endings are "Windows (CRLF)", and the character set is "UTF-8".

Terraform Destroy

```
C:\Users\Manas\Documents\exercise5>
C:\Users\Manas\Documents\exercise5>terraform state list
C:\Users\Manas\Documents\exercise5>terraform destroy
No changes. No objects need to be destroyed.

Either you have not created any objects yet or the existing objects were already deleted outside of Terraform.

Destroy complete! Resources: 0 destroyed.
C:\Users\Manas\Documents\exercise5>
```

